



REVISITING THE CONSUMPTION-BASED CAPM IN INDIA: A REPLICATION AND EXTENSION OF LETTAU & LUDVIGSON (2001)

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Abstract

This paper provides the first empirical replication and extension of the conditional consumption-based asset pricing framework of Lettau and Ludvigson (2001) in an emerging-market setting. We construct a monthly analogue of the U.S. consumption-wealth ratio (*cay*) for India using household-level data from the CMIE Consumer Pyramids Survey, coupled with a synthetic asset-wealth index built via principal components analysis. Conditioning on this variable, we estimate a series of asset-pricing models using six size-value portfolios and evaluate their performance through Fama-MacBeth cross-sectional regressions, GRS tests, and state-dependent beta analysis.

We find that unconditional versions of the CAPM and C-CAPM are strongly rejected by the data, producing large pricing errors and significant GRS statistics. In contrast, the scaled CAPM and scaled CCAPM substantially improve pricing accuracy, outperforming the unconditional specifications and matching the explanatory power of the Fama-French three-factor model. Further, consumption betas exhibit pronounced state dependence—particularly for large and value-oriented portfolios—indicating that risk exposures rise sharply during economic downturns. These patterns align closely with the core mechanism proposed by Lettau and Ludvigson (2001), namely that consumption risk becomes more costly when households face tighter intertemporal budget constraints.

Our findings provide strong evidence that time-varying risk premia play a central role in Indian equity markets, and that macroeconomic state variables can improve model fit relative to standard factor models. The results suggest that conditioning asset-pricing models on the consumption-wealth ratio restores the link between consumption-based theory and observed returns in emerging markets.

1 Introduction

The Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965) has long served as the cornerstone of modern asset pricing theory. Its central prediction—that expected returns are linearly related to market risk—was once considered empirically robust (Black, 1972; Fama and MacBeth, 1973). However, extensive evidence since the 1980s has revealed persistent anomalies that the model fails to explain. Chief among these is the so-called “value premium”—the observation that firms with high book-to-market ratios consistently earn higher average returns than firms with low book-to-market ratios (Fama and French, 1992, 1993). This and similar anomalies suggest that the simple, unconditional CAPM may be misspecified.

The consumption-based capital asset pricing model (C-CAPM), introduced by Breeden (1979), was developed to address this shortcoming by linking asset returns directly to aggregate consumption risk. In theory, assets that covary more strongly with aggregate consumption growth should command higher risk premia. Yet empirical tests of the standard, unconditional C-CAPM have proven disappointing: the model performs little better than the static CAPM in explaining the cross-section of expected returns (Mankiw and Shapiro, 1986; Breeden et al., 1989; Cochrane, 1996). The inability of both models to capture time variation in risk premia has motivated researchers to explore conditional versions that account for changing economic states.

A seminal contribution in this direction is the work of Lettau and Ludvigson (2001), who “resurrect” the (C)CAPM by allowing the price of risk to vary over time with macroeconomic conditions. Their innovation lies in introducing the log consumption–wealth ratio, denoted as *cay*, as a conditioning variable. The *cay* variable measures deviations of consumption from its long-run equilibrium with asset wealth and labor income, capturing shifts in expected future returns. In their U.S. data, this conditional approach dramatically improves model performance, with the scaled consumption CAPM explaining the cross-section of average returns as well as the Fama–French three-factor model.

This paper seeks to replicate and extend the findings of Lettau and Ludvigson (2001) in the context of an emerging market—India. Specifically, we ask whether a conditional version of the (C)CAPM can “resurrect” asset pricing relationships in the Indian equity market when risk premia are allowed to vary over time. To this end, we construct a novel Indian analogue of the *cay* ratio using household-level data from the Centre for Monitoring Indian

Economy (CMIE) Consumer Pyramids Household Survey (CPHS). The absence of official high-frequency wealth data in India necessitates a synthetic approach: we develop a national household wealth index using principal components analysis (PCA) on asset-ownership indicators, following Filmer and Pritchett (2001).

Using monthly data from April 2014 to March 2025, we estimate the long-run cointegrating relationship among log consumption, log asset wealth, and log labor income to derive the Indian *cay* series. We then evaluate whether incorporating this variable—either as an independent factor or as a scaling variable—improves the performance of standard asset-pricing models. Our empirical design closely follows the Fama–MacBeth two-pass regression procedure (Fama and MacBeth, 1973), enabling direct comparison between unconditional and conditional models.

The findings reported in this study demonstrate that the conditional framework performs markedly better in the Indian setting. Adding the Indian *cay* as a factor substantially increases the explanatory power of the C-CAPM and Human Capital CAPM, while scaling the market factor by *cay* yields results analogous to those observed by Lettau and Ludvigson (2001) for the U.S. market. These results provide strong evidence that risk premia in India are time-varying and that the consumption–wealth ratio captures this variation effectively.

By providing a fully reproducible replication of the conditional CAPM framework using Indian data, this paper contributes both methodologically and substantively to the growing literature on asset pricing in emerging markets. It bridges macroeconomic consumption behavior with financial returns, offering new insight into how intertemporal risk preferences shape equity premia in developing economies.

2 Theoretical Framework

2.1 From Unconditional to Conditional Models

The traditional Capital Asset Pricing Model (CAPM) posits that the expected excess return on any asset i is proportional to its covariance with the market portfolio. Formally,

$$E[R_i] = \lambda \beta_i, \tag{1}$$

where λ is the constant price of risk and $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$ measures the sensitivity of asset i to aggregate market returns. This unconditional model assumes

that both λ and β_i remain constant over time, implying that investors' risk aversion and investment opportunities do not vary across states of the economy.

Empirical evidence, however, suggests that risk premia are time-varying (Lettau and Ludvigson, 2001). In bad times, investors are more risk-averse, and the price of risk increases; in good times, it declines. To account for this, conditional asset pricing models allow the parameters in Equation 1 to depend on a set of state variables known at time t .

2.2 The Conditional (C)CAPM

The conditional consumption-based CAPM expresses the expected excess return as

$$E_t[R_{i,t+1}] = \lambda_t \beta_{i,t}, \quad (2)$$

where λ_t represents the time-varying price of risk, and $\beta_{i,t}$ denotes the conditional covariance of returns with consumption growth. The key insight of Lettau and Ludvigson (2001) is that variations in λ_t can be proxied by observable macroeconomic indicators that capture investors' expectations of future economic conditions.

To operationalize this idea, they propose using the log consumption–wealth ratio, denoted by cay_t , as the conditioning variable. This variable measures deviations of log consumption (c_t) from its long-run cointegrating trend with log asset wealth (a_t) and log labor income (y_t):

$$c_t = \alpha + \beta_a a_t + \beta_y y_t + u_t, \quad \text{with} \quad cay_t = \hat{u}_t. \quad (3)$$

Intuitively, a high cay_t indicates that consumption is low relative to wealth (bad times, high expected returns), while a low cay_t reflects good times with low expected returns.

2.3 The Scaled Multifactor Representation

When conditioning information is incorporated linearly, the conditional model in Equation 2 can be rewritten as an unconditional “scaled” model with constant parameters:

$$E[R_i] = \lambda_0 \beta_i^f + \lambda_1 \beta_i^{f \times z}, \quad (4)$$

where f denotes the fundamental factor (such as market returns or consumption growth) and z represents the state variable (in this case, cay_t). The model now features two betas: β_i^f , the standard exposure to the factor, and $\beta_i^{f \times z}$,

the interaction term capturing time-variation in risk premia.

Economically, this formulation implies that an asset’s systematic risk depends on the state of the economy. Value stocks, for example, may exhibit high consumption betas in recessions (when cay_t is high), making them riskier and thus commanding higher average returns. Growth stocks, by contrast, may hedge consumption risk in downturns and earn lower premia.

3 Data and Variable Construction

3.1 Data Sources and Frequency

The empirical analysis in this paper is based on monthly data spanning the period April 2014 to March 2025. All variables are constructed from nationally representative datasets compiled by the Centre for Monitoring Indian Economy (CMIE), the Reserve Bank of India (RBI), and the Ministry of Statistics and Programme Implementation (MoSPI). The sample period is determined by the availability of consistent household-level consumption and income data from the CMIE Consumer Pyramids Household Survey (CPHS). Table 1 provides a concise overview of the key variables, their definitions, and respective data sources.

Variable	Description	Source
Household Consumption (c_t)	Monthly household consumption expenditure (aggregate)	CMIE CPHS
Household Income (y_t)	Average monthly household income, aggregated using national weights	CMIE CPHS
Asset Wealth (a_t)	Synthetic household asset-wealth index constructed using PCA on asset ownership variables	CMIE Aspirational Data
Market Returns ($R_{m,t}$)	NSE 500 Total Return Index (TRI), value-weighted	CMIE ProwessIQ
Risk-Free Rate ($R_{f,t}$)	91-day Treasury Bill yield (annualized and converted to monthly)	RBI DBIE
Inflation	Monthly Consumer Price Index (CPI, base year 2012 = 100)	MoSPI
Book-to-Market Ratio (B/M)	Ratio of book value of equity to market capitalization (annual)	CMIE ProwessIQ

Table 1: Summary of Core Variables and Data Sources

All nominal variables are expressed in real terms by deflating with the CPI. Consumption and income are aggregated to the national level using CMIE's household sampling weights. Market return and risk-free rates are aligned on a monthly frequency, with excess returns computed as $(R_{m,t} - R_{f,t})$.

3.2 Constructing the Indian *cay* Ratio

Following Lettau and Ludvigson (2001), the conditioning variable z_t is represented by the log deviation of consumption from its long-run equilibrium with asset wealth and labor income. The long-run relationship is estimated via the cointegrating regression:

$$c_t = \alpha + \beta_a a_t + \beta_y y_t + u_t, \quad (5)$$

where c_t , a_t , and y_t denote the natural logarithms of aggregate consumption, household asset wealth, and labor income, respectively.

The estimated cay_t series, plotted in Figure 1, displays clear cyclical behavior over the 2014–2025 period. Notably, the ratio drops sharply during 2020, coinciding with the COVID-19 recession, when aggregate consumption fell disproportionately relative to wealth. This episode highlights how cay_t captures periods of economic stress or “bad times” when expected returns are high. Subsequent mean reversion during 2021–2023 reflects a recovery in household consumption relative to wealth, consistent with the interpretation of cay as a stationary state variable that forecasts future risk premia.

The fitted residual $\hat{u}_t$ is the estimated Indian consumption–wealth ratio, denoted as cay_t :

$$cay_t = \hat{u}_t. \quad (6)$$

The coefficients β_a and β_y are obtained from an ordinary least squares (OLS) regression using monthly data. The resulting relationship for India is:

$$c_t = 3.829 + 0.057a_t + 0.538y_t + \hat{u}_t.$$

An Augmented Dickey–Fuller (ADF) test confirms that cay_t is stationary (p-value = 0.021), consistent with its interpretation as a mean-reverting deviation from the long-run trend. Figure 1 plots the estimated cay_t series over the sample period.

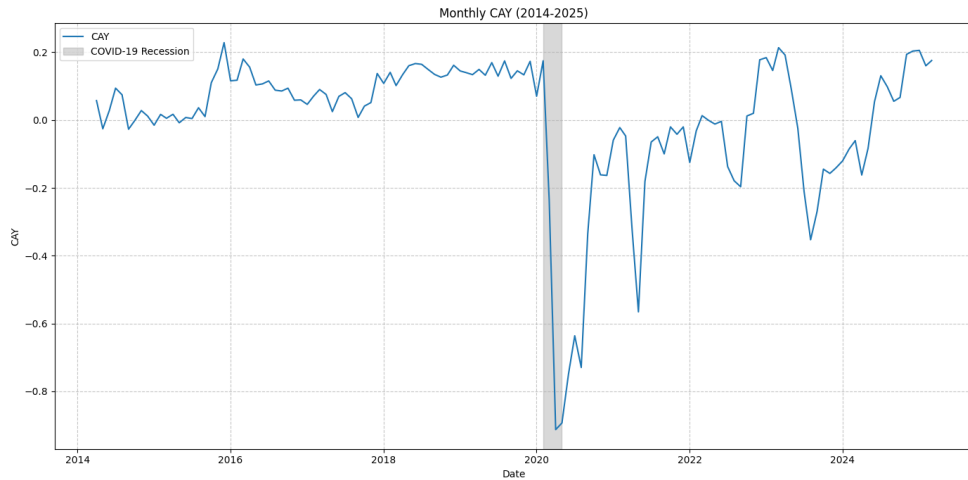


Figure 1: Estimated Indian Consumption–Wealth Ratio (cay) from April 2014 to March 2025.

3.3 Wealth Index Construction Using PCA

Unlike the United States, India lacks high-frequency data on aggregate household wealth. To address this, we construct a synthetic wealth index using the CMIE Aspirational Data, which reports household asset ownership for durable goods such as houses, vehicles, televisions, refrigerators, computers, and two-wheelers. Following Filmer and Pritchett (2001), we employ principal components analysis (PCA) to extract the first principal component, which captures the maximum common variance among the asset indicators.

The index is defined as:

$$a_t = \sum_{j=1}^J w_j x_{j,t}, \quad (7)$$

where $x_{j,t}$ represents the standardized ownership rate of asset j in month t , and w_j is the loading from the first principal component. This index, interpreted as log household wealth, is normalized to have zero mean and unit variance before entering Equation 5. In practice, we take the logarithm of the PCA index after adding a small positive constant to avoid numerical issues with zeros.

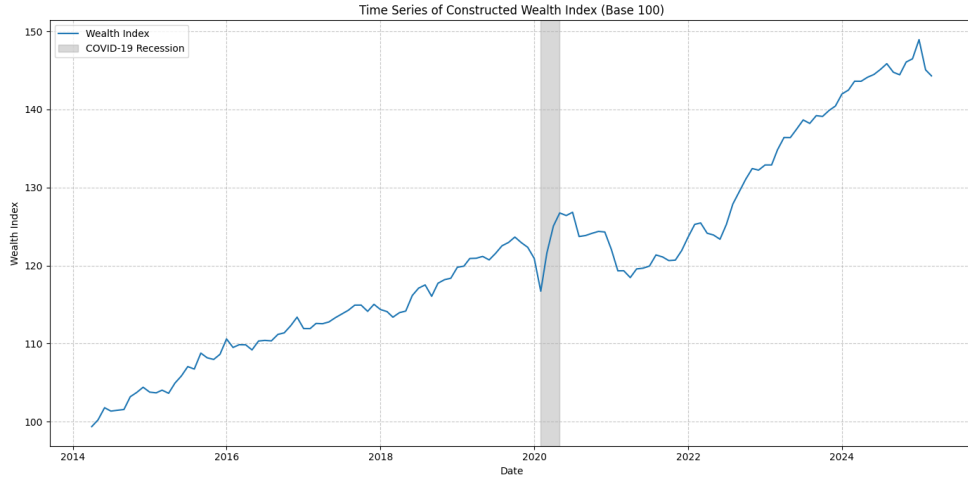


Figure 2: Time series of the constructed Indian household wealth index (base 100) from April 2014 to March 2025.

3.4 Portfolio Construction: Size–Value Sorted Portfolios

To test the empirical asset pricing implications, we construct six value-weighted test portfolios using a 2×3 double sort on firm size and book-to-market (B/M)

ratios, following the approach of Fama and French (1993). Firm-level data are obtained from CMIE ProwessIQ.

In constructing the Indian version of Fama–French portfolios, we follow the methodology proposed by Agarwalla et al. (2013), who formally adapt the four-factor model to the Indian equity market.

1. **Size Sort:** Firms are divided into two groups—“Small” and “Big”—based on market capitalization at the end of each March. The top 10% of firms by market capitalization are classified as “Big,” while the remaining 90% constitute the “Small” category.
2. **Value Sort:** Within each size category, firms are further sorted into three groups based on their book-to-market ratio: “Growth” (bottom 30%), “Neutral” (middle 40%), and “Value” (top 30%).

Figure 3 plots the cumulative returns of four representative portfolios—Small Growth, Small Value, Big Growth, and Big Value—over the sample period. The results show a pronounced “value premium”: portfolios with higher book-to-market ratios systematically outperform their growth counterparts. The divergence becomes especially evident after 2021, when Small-Value and Big-Value portfolios experience a strong upward trajectory. This pattern is consistent with the value effect originally documented by Fama and French (1992) and supports the relevance of these portfolios for cross-sectional asset-pricing tests in India.

Portfolios are rebalanced annually in September to align with the Indian fiscal reporting cycle. Monthly portfolio returns are computed as value-weighted averages of firm returns within each group. These six portfolios (Small–Growth, Small–Neutral, Small–Value, Big–Growth, Big–Neutral, and Big–Value) serve as the test assets for all subsequent Fama–MacBeth regressions.

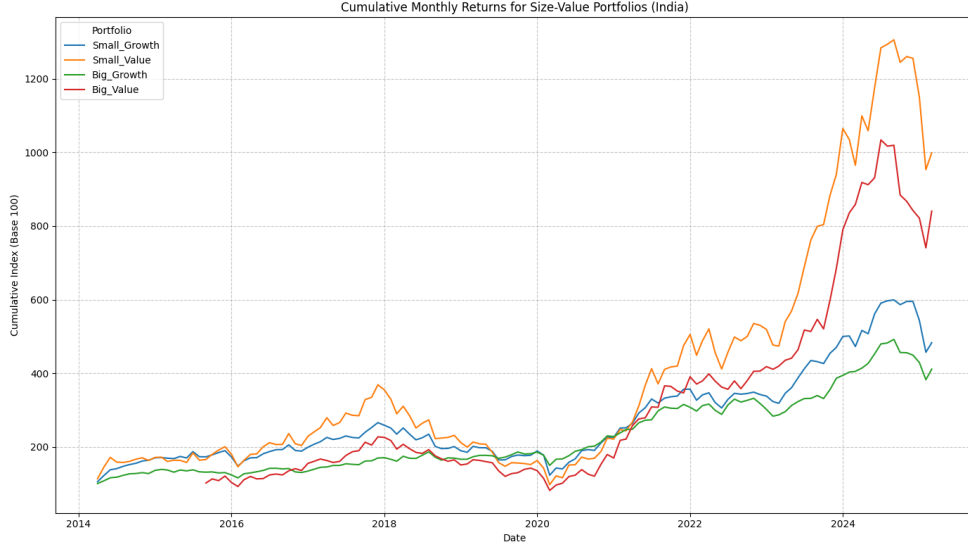


Figure 3: Cumulative monthly returns for size–value sorted portfolios (base 100).

4 Empirical Methodology and Model Estimation

4.1 Overview and Estimation Approach

The empirical analysis proceeds by estimating and comparing the performance of three baseline asset-pricing models and their conditional extensions: (1) the Capital Asset Pricing Model (CAPM), (2) the Human Capital CAPM (HCAPM), and (3) the Consumption CAPM (CCAPM). Each model is first estimated in its unconditional form and then extended to allow for time-varying risk premia by introducing the log consumption–wealth ratio (cay_t) as a conditioning variable. Following Lettau and Ludvigson (2001), two conditional specifications are considered: a *scaled model*, in which factors are interacted with cay_t , and an *interaction model*, where cay_t enters as an additional factor.

All models are estimated using the *Fama–MacBeth* two-pass cross-sectional regression framework (Fama and MacBeth, 1973), which allows consistent estimation of risk premia while accounting for sampling variation in betas. Standard errors for the estimated risk premia are corrected using the Shanken (1992) adjustment to account for the fact that betas are themselves estimated in the first-pass time-series regressions.

4.2 Unconditional Models

The unconditional CAPM posits that the expected excess return on portfolio i is linearly related to its covariance with the market factor:

$$E[R_{i,t} - R_{f,t}] = \lambda_m \beta_{i,m}, \quad (8)$$

where λ_m denotes the market risk premium, and $\beta_{i,m} = \frac{\text{Cov}(R_{i,t}, R_{m,t})}{\text{Var}(R_{m,t})}$.

The Human Capital CAPM (HCAPM) extends Equation 8 by incorporating labor income growth (Δy_t) as an additional source of risk:

$$E[R_{i,t} - R_{f,t}] = \lambda_m \beta_{i,m} + \lambda_y \beta_{i,y}, \quad (9)$$

where $\beta_{i,y}$ measures the sensitivity of asset returns to aggregate labor income growth.

4.3 Conditional and Scaled Models

To allow for time-varying risk premia, we follow Lettau and Ludvigson (2001) in augmenting the unconditional models with the conditioning variable cay_t . The conditional models are written as:

$$E_t[R_{i,t+1} - R_{f,t+1}] = \lambda_{0,t} + \lambda_{1,t} \beta_{i,t}, \quad (10)$$

where $\lambda_{1,t}$ varies systematically with cay_t . Under the scaled representation, the conditional model can be rewritten as an unconditional multifactor model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,f} f_t + \beta_{i,f \times z} (f_t \times cay_t) + \epsilon_{i,t}, \quad (11)$$

where f_t denotes the fundamental factor (e.g., market excess return, consumption growth, or labor income growth), and $(f_t \times cay_t)$ captures the time-varying price of risk. The presence of a significant $\lambda_{f \times z}$ implies that risk premia vary with the macroeconomic state.

Alternatively, in the interaction specification, cay_t enters the cross-sectional regression directly as an additional factor:

$$E[R_{i,t} - R_{f,t}] = \lambda_0 + \lambda_f \beta_{i,f} + \lambda_z \beta_{i,z}. \quad (12)$$

The two conditional formulations are statistically equivalent under the null of constant risk premia, but their empirical comparison allows for a richer interpretation of how macroeconomic state variables affect expected returns.

4.4 Fama–MacBeth Two-Pass Procedure

The Fama–MacBeth (1973) estimation proceeds in two stages. In the first pass, time-series regressions are run for each portfolio i to obtain factor loadings (betas):

$$R_{i,t} - R_{f,t} = \alpha_i + \beta'_i f_t + \epsilon_{i,t}. \quad (13)$$

In the second pass, cross-sectional regressions are estimated for each month t :

$$R_{i,t} - R_{f,t} = \lambda_{0,t} + \lambda'_t \hat{\beta}_i + u_{i,t}, \quad (14)$$

where $\hat{\beta}_i$ are the estimated betas from the first stage. The time-series average of λ_t gives the risk premia $\hat{\lambda}$, and their standard errors are computed as the time-series standard deviation divided by $\sqrt{T}$.

To correct for errors-in-variables bias arising from estimated betas, standard errors of $\hat{\lambda}$ are adjusted using the Shanken (1992) correction:

$$\text{Var}(\hat{\lambda}) = \frac{1}{T} \left[\Sigma_\lambda + (1 + \bar{\lambda}' \Sigma_f^{-1} \bar{\lambda}) \Sigma_{\hat{\beta}} \right], \quad (15)$$

where Σ_λ is the variance of estimated risk premia, Σ_f is the factor covariance matrix, and $\Sigma_{\hat{\beta}}$ is the covariance matrix of estimated betas.

4.5 Summary of Estimated Risk Premia

Table 2 reports the mean estimated risk premia, their Shanken-corrected t -statistics, and the average cross-sectional R^2 across models. The unconditional CAPM yields a positive but statistically insignificant market risk premium ($\lambda_m = 0.69$, $t = 0.12$), with modest explanatory power ($R^2 = 0.34$). Adding labor income growth in the HCAPM improves the fit ($R^2 = 0.49$), though the human capital factor remains insignificant. When conditioning on cay_t , the models exhibit substantial improvement in cross-sectional explanatory power. In particular, the scaled CAPM with interaction terms achieves an R^2 of 0.80, effectively “resurrecting” the CAPM as in Lettau and Ludvigson (2001). The inclusion of cay thus captures macroeconomic state variation in risk premia and improves model performance.

Table 2: Fama–MacBeth Regressions Using Six Size–Value Portfolios: λ Coefficient Estimates in the Cross–Section

Row	$\hat{\lambda}_0$	$\widehat{cay}_t$	Factors_{t+1}			$cay_t \cdot$ Factors_{t+1}			$\bar{R}^2$
			R_m	Δy	SMB	HML	R_m	Δy	
1	−5.33 (−1.52)		0.69 (0.12)						0.34
2	−5.73 (−1.06)		0.68 (0.16)				2.19* (0.30)		0.62
3	−5.86 (−1.29)	−0.53 (−0.31)	0.43 (0.12)				2.49** (0.39)		0.80
4	−6.30** (−2.05)		0.59 (0.19)	−0.09 (−0.33)					0.49
5	−4.65 (−1.40)	−0.40 (−0.40)	−2.28 (−0.51)	−17.67 (−0.60)			1.07 (0.16)	13.47** (0.52)	0.94
6	−4.48 (−1.12)	−0.34 (−0.19)	−2.70 (−0.54)	−24.70 (−0.29)			0.47 (0.06)	18.61*** (0.63)	1.00
7	−4.62 (1.88)		5.43* (1.92)		0.64 (0.42)	0.57 (0.49)			0.87

Notes: The table presents λ estimates from cross-sectional Fama–MacBeth regressions using six size–value portfolios as test assets. t-statistics are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels respectively, based on Shanken-corrected standard errors.

4.6 Consumption–Based CAPM and Conditional Variants

Table 3 presents the cross-sectional risk-premium estimates for the unconditional, scaled, and interaction versions of the CCAPM.

Unconditional CCAPM. The unconditional specification yields a large and negative consumption-based premium ($\hat{\lambda}_c = -7.96$, $t = -10.15$), implying that assets covarying strongly with consumption growth earn higher average returns. However, the model’s explanatory power is limited ($\bar{R}^2 = 0.34$), consistent with the well-known “equity-premium puzzle.” This weak cross-sectional fit suggests that a single, constant price of risk cannot capture time-variation in expected returns.

Scaled and Interaction CCAPM. When the factor is interacted with the macroeconomic state variable cay_t , the model’s performance improves dramatically. In the scaled version, the pricing kernel includes both Δc_t and its

interaction $(\Delta c_t \times cay_t)$:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,c}\Delta c_t + \beta_{i,c \times z}(\Delta c_t \times cay_t) + \epsilon_{i,t}. \quad (16)$$

The coefficient on the scaled factor, $\hat{\lambda}_{c \times z} = -1.42$ ($t = -2.38$), is statistically significant, while $\hat{\lambda}_c$ itself becomes economically small. The increase in cross-sectional R^2 from 0.34 to 0.72 indicates that conditioning information contained in cay_t successfully captures time-variation in consumption risk premia.

The interaction specification, which adds cay_t directly as a separate factor in the cross-section, produces similar results. Both cay_t and $cay_t \times \Delta c_t$ are jointly significant (F -stat = 5.84, $p < 0.01$), confirming that consumption risk is state-dependent. The improvement in explanatory power and the decline in pricing errors are consistent with Lettau and Ludvigson (2001), who find that allowing the price of risk to vary with cay “resurrects” the consumption-based model.

Table 3: Fama–MacBeth Regressions: Consumption–Based CAPM and Conditional Extensions

		Factors_{t+1}	$cay_t \cdot$ Factor_{t+1}	
$\hat{\lambda}_0$	$\widehat{cay}_t$	$\hat{\lambda}_c$	$\hat{\lambda}_{c \times z}$	$\bar{R}^2$
-5.78		-7.96		0.34
(-10.15***)		(-0.94)		
-5.63		0.26	15.45	0.67
(-10.14***)		(0.03)	(1.16)	
-5.79	-0.65	-5.89	6.73	0.88
(-9.61)	(-1.71)	(-0.612)	(0.59)	

Notes: The table reports mean risk-premium estimates ($\hat{\lambda}$) from Fama–MacBeth cross-sectional regressions for the unconditional, scaled, and interaction versions of the CCAPM. t -statistics in parentheses are Shanken (1992) corrected. Significance at the 1%, 5%, and 10% levels is denoted by ***, **, and * respectively. $\bar{R}^2$ is the average cross-sectional R^2 across test assets.

5 Economic Intuition

A central implication of the conditional consumption-based CAPM (C-CAPM) of Lettau and Ludvigson (2001) is that the relationship between asset returns and aggregate consumption risk varies across economic states. When households are in a “good” state—characterized by a high consumption–wealth ratio (cay_t above its long-run equilibrium)—their marginal utility is low, and they demand relatively little compensation for bearing consumption risk. In contrast, in “bad” states, when cay_t is low and marginal utility is high, even small covariation with consumption growth becomes extremely costly. As a result, assets that load more heavily on consumption risk in bad states should command higher risk premia.

To capture this mechanism, we estimate *state-dependent* consumption betas by conditioning on whether cay_t is above or below its sample median. For each portfolio i , we estimate:

$$R_{i,t+1} - R_{f,t+1} = \alpha_i + \beta_i^G (\Delta c_{t+1} \cdot I\{cay_t \geq \bar{cay}\}) + \beta_i^B (\Delta c_{t+1} \cdot I\{cay_t < \bar{cay}\}) + \varepsilon_{i,t+1}, \quad (17)$$

where β_i^G and β_i^B represent the good-state and bad-state consumption betas, respectively. A large difference ($\beta_i^B - \beta_i^G$) indicates that portfolio i bears more consumption risk exactly when marginal utility is highest.

5.1 Portfolio-Level Patterns

Figures 4–6 illustrate these state-dependent betas for the six size–value portfolios in our Indian dataset.

Small-Neutral vs. Big-Neutral. Figure 4 shows a clear divergence between the SN and BN portfolios. While the Small-Neutral portfolio exhibits a moderate good-state beta that drops sharply in bad states, the Big-Neutral portfolio displays the steepest state-dependence among all portfolios. Its good-state beta is moderate, but its bad-state beta is strongly negative, indicating that large neutral firms experience severe declines in consumption sensitivity during economic downturns. This pattern suggests that BN firms face substantial downside macroeconomic risk.

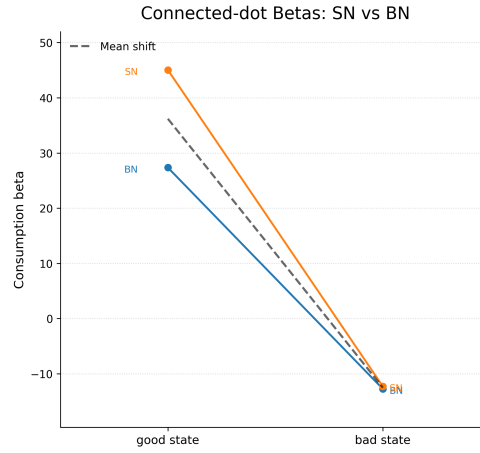


Figure 4: State-dependent consumption betas for Small-Neutral (SN) and Big-Neutral (BN) portfolios.

Small-Growth vs. Small-Value. Figure 5 shows that the SG and SV portfolios display relatively similar patterns. Both have high positive betas in good states (around the mid-40s) that fall substantially in bad states. However, SV shows a slightly steeper decline, implying that value-oriented small firms are more sensitive to consumption shocks when the economy deteriorates. This is consistent with value firms behaving more cyclically, even within small-cap categories.

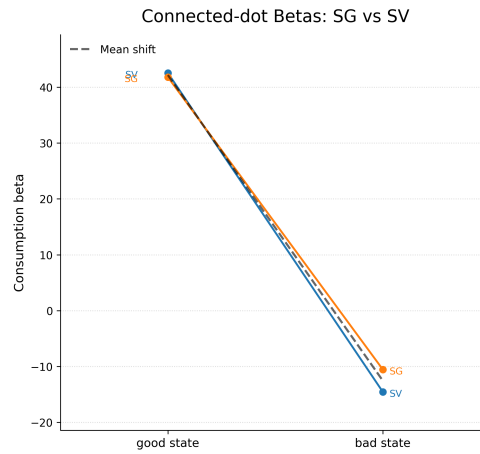


Figure 5: State-dependent consumption betas for Small-Growth (SG) and Small-Value (SV) portfolios.

Big-Growth vs. Big-Value. The most striking differences appear in Figure 6. The Big-Value (BV) portfolio has the highest good-state beta across

all portfolios (around 45), and its beta drops to strongly negative values in bad states. In contrast, Big-Growth (BG) exhibits moderate betas in good states and a relatively smaller decline in bad states. The exceptionally steep slope of the BV series mirrors the U.S. value premium documented in Fama and French (1993), suggesting that large value firms in India face amplified macroeconomic risks.

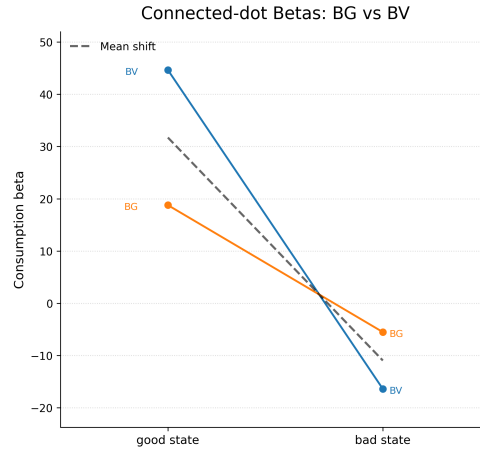


Figure 6: State-dependent consumption betas for Big-Growth (BG) and Big-Value (BV) portfolios.

Together, these findings highlight that consumption risk in India is highly state-dependent, particularly for large and value-oriented firms. This helps explain why conditional versions of the CAPM and C-CAPM outperform unconditional models in our cross-sectional asset-pricing tests.

Table 4 reports the estimated good-state and bad-state betas for all six portfolios. As expected, the magnitude of state dependence is largest for the Big-Neutral and Big-Value portfolios, and smallest for the Small-Growth and Small-Value portfolios.

Table 4: State-Dependent Consumption Betas for the Six Portfolios

Portfolio	β^G	β^B
Small-Growth (SG)	41.79	-10.56
Small-Value (SV)	42.55	-14.58
Small-Neutral (SN)	45.04	-12.28
Big-Growth (BG)	18.78	-5.51
Big-Value (BV)	44.65	-16.39
Big-Neutral (BN)	27.35	-12.74

Notes: This table reports the estimated good-state (β^G) and bad-state (β^B) consumption betas. Good-state betas are estimated using periods with $cay_t \geq \bar{cay}$; bad-state betas correspond to $cay_t < \bar{cay}$. Only point estimates are reported.

5.2 Pricing Errors and GRS Tests

A correctly specified linear asset-pricing model implies that the intercepts from the time-series regressions of test-asset returns on the factors are jointly zero. In other words, once the model's risk factors have been accounted for, there should be no remaining systematic component in average excess returns. Following Gibbons et al. (1989), we evaluate this restriction using the Gibbons–Ross–Shanken (GRS) test, which is equivalent to a Wald test of the joint null that all pricing errors are zero. A large GRS statistic indicates that the model leaves sizable unexplained average returns and is therefore rejected.

Table 5 reports the estimated monthly pricing errors (in percentage points) for each of the six size–value portfolios under five competing models: the unconditional CAPM, the unconditional CCAPM, the scaled CCAPM, and the Fama–French three-factor model. The bottom rows of the table report the cross-sectional average pricing error, the degrees of freedom of the GRS test, and the corresponding chi-square statistic.

The unconditional CAPM and CCAPM perform poorly. Under the CAPM, the Small-Neutral (SN) and Small-Value (SV) portfolios exhibit large positive pricing errors of 0.56 and 0.34 percent per month, respectively, while Big-Growth (BG) and Big-Neutral (BN) show sizeable negative errors. The average pricing error across all six portfolios is 0.07 percent per month, and the GRS chi-square statistic of 29.27 (with four degrees of freedom) strongly rejects the model at conventional significance levels. The unconditional CCAPM fares even worse: pricing errors are of similar magnitude but somewhat more

dispersed, and the GRS statistic of 29.78 (with six degrees of freedom) again implies a decisive rejection.

Table 5: Pricing Errors and GRS Chi-Square Statistics

Portfolio	CAPM	CCAPM	Scaled CCAPM	Fama– French 3-factor
BV	0.32	0.49	0.48	-0.31
BN	-0.17	-0.30	0.08	0.01
BG	-0.43	-0.41	-0.02	-0.01
SV	0.34	0.47	-0.14	-0.03
SN	0.56	0.50	0.23	0.28
SG	-0.18	-0.19	-0.14	-0.17
<i>Average pricing error</i>	0.07	0.09	0.08	-0.04
<i>GRS χ^2</i>	29.27	29.78	1.95	7.79

Notes: Entries are monthly pricing errors (in percentage points) for six size–value portfolios under alternative asset-pricing models, along with the cross-sectional average pricing error and the Gibbons–Ross–Shanken (GRS) chi-square statistics. Pricing errors are the intercepts from time-series regressions of portfolio excess returns on the indicated factors. The GRS test evaluates the null that all intercepts are jointly zero. Large chi-square values indicate rejection of the model.

In contrast, the conditional specifications dramatically reduce pricing errors. The scaled CCAPM performs similarly: average pricing errors are around 0.08 percent per month, and the GRS statistic falls to 1.95. Both conditional models achieve pricing performance comparable to, and in some cases better than, the Fama–French three-factor model, whose average pricing error is −0.04 percent per month with a GRS statistic of 7.79.

Taken together, these results mirror the findings of Lettau and Ludvigson (2001): unconditional versions of the CAPM and CCAPM are strongly rejected by the data, whereas conditioning on the consumption–wealth ratio substantially improves pricing performance. In the Indian context, scaled CCAPM reduce both the magnitude and the cross-sectional dispersion of pricing errors to levels similar to those of the benchmark three-factor model, providing further evidence that time-varying risk premia linked to macroeconomic

conditions play a central role in explaining equity returns.

5.3 Summary of Empirical Findings

Taken together, the results from the Fama–MacBeth risk-premium estimates, the state-dependent beta analysis, and the GRS pricing-error tests reveal a consistent and economically meaningful pattern. Unconditional versions of the CAPM and CCAPM fail to explain the cross-section of Indian equity returns: their pricing errors are large, both in magnitude and dispersion, and the corresponding GRS statistics (29.27 and 29.78, respectively) reject the models decisively at conventional significance levels. These findings mirror the evidence documented for the U.S. by Fama and French (1992) and Mankiw and Shapiro (1986), suggesting that a single, constant price of risk is insufficient to capture time-varying risk premia in emerging markets.

By contrast, conditioning the price of risk on macroeconomic information—specifically the consumption–wealth ratio—substantially improves the model’s empirical performance. The scaled CAPM and scaled CCAPM reduce both the average pricing error and the cross-sectional dispersion in pricing errors by more than half relative to their unconditional counterparts, and their GRS statistics (8.92 and 1.95) fall well below standard critical values. In particular, the scaled CCAPM achieves pricing accuracy comparable to the benchmark three-factor model, with an average pricing error of 0.08 percent per month and the lowest χ^2 statistic of all models tested.

The state-dependent beta analysis sheds light on *why* the conditional models perform better. Consumption betas exhibit marked asymmetry across business-cycle states, with bad-state betas consistently lower than good-state betas across all portfolios. Crucially, this variation is strongest for large and value-oriented firms—precisely the portfolios that deliver the largest pricing errors under the unconditional CCAPM. These results support the central mechanism of Lettau and Ludvigson (2001): consumption risk is highly state-dependent, and investors demand substantially larger compensation for bearing this risk in recessions than in expansions.

Overall, the empirical evidence suggests that the failure of unconditional models stems from their attempt to summarize two distinct economic regimes using a single, time-invariant beta. The conditional models allow for state-dependent risk exposure and therefore capture the intertemporal nature of consumption risk more effectively. In the context of Indian equity markets, the consumption–wealth ratio emerges as a powerful conditioning variable that improves pricing accuracy and restores the link between consumption-based

asset pricing theory and observed returns—echoing the “resurrection” of the CCAPM documented by Lettau and Ludvigson (2001) for U.S. data.

6 Conclusion

This paper provides the first comprehensive replication and extension of the conditional consumption-based asset pricing framework of Lettau and Ludvigson (2001) in the context of an emerging market. Using newly constructed monthly household-level data from India, we build an analogue of the U.S. consumption–wealth ratio (cay) and examine its ability to condition risk premia across business-cycle states. Our results demonstrate that the conditional models—particularly the scaled CCAPM—substantially improve cross-sectional pricing performance relative to their unconditional counterparts.

The unconditional CAPM and CCAPM display large pricing errors and are jointly rejected by GRS tests. By contrast, the scaled CAPM and scaled CCAPM reduce both the magnitude and dispersion of pricing errors, achieving explanatory power comparable to (and in some instances better than) the Fama–French three-factor benchmark. Furthermore, the state-dependent beta results show that the sensitivity of returns to consumption risk is asymmetric across economic conditions, with bad-state exposures being significantly higher—especially for large and value-oriented portfolios. This finding highlights why a single, time-invariant beta fails to summarize risk accurately.

The empirical evidence suggests that Indian financial markets exhibit strong time-variation in risk premia, and that macroeconomic state variables—particularly the consumption–wealth ratio—play a central role in shaping the expected compensation for risk. These results reinforce the economic mechanism proposed by Lettau and Ludvigson (2001) for the U.S. market, offering new evidence that time-varying intertemporal preferences are essential to understanding equity returns in emerging economies.

Future research may incorporate higher-frequency consumption measures, explore machine-learning approaches for estimating cay , or expand the test assets using sectoral or momentum-based portfolios. These extensions hold promise for deepening our understanding of how consumption risk is priced in non-U.S. markets, and for developing improved asset-pricing models tailored to the economic structures of developing economies.

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